

Smart Agriculture Project: Code Explanation

This document explains the Arduino code for the Smart Agriculture Project, an IoT-based system for intelligent irrigation control using an ESP8266 NodeMCU. The system automates irrigation based on soil moisture, weather forecasts, and time of day, with manual control via the Blynk mobile app. It integrates with ThingSpeak for data logging and weather data retrieval.

1. Overview

The code runs on an ESP8266 NodeMCU microcontroller and performs the following tasks:

- Reads soil moisture levels using an analog sensor.
- Fetches weather forecasts from ThingSpeak to check for expected rain.
- Synchronizes time via NTP servers to schedule irrigation during optimal hours (5:00–9:00 AM, 6:00–8:00 PM).
- Controls a water pump via a relay based on automated logic or manual input from Blynk.
- Logs soil moisture data to ThingSpeak for remote monitoring.
- Uses LEDs to indicate system status (pump, rain, and general status).

2. Code Structure

The code is written in C++ for the Arduino platform and uses libraries for Wi-Fi, Blynk, HTTP requests, and time synchronization. It is organized into setup, loop, and helper functions, with a Blynk callback for manual control.

2.1 Libraries and Definitions

- **Libraries:**

- `ESP8266WiFi.h`: Enables Wi-Fi connectivity for the ESP8266.
- `BlynkSimpleEsp8266.h`: Facilitates communication with the Blynk mobile app.
- `ESP8266HTTPClient.h`: Handles HTTP requests to ThingSpeak.
- `time.h`: Provides time synchronization via NTP.

- **Blynk Configuration:**

- `BLYNK_TEMPLATE_ID`, `BLYNK_TEMPLATE_NAME`, `BLYNK_AUTH_TOKEN`: Credentials for connecting to the Blynk server, allowing remote control and monitoring.

- **Wi-Fi Configuration:**

- `ssid` and `password`: Credentials for the local Wi-Fi network (`TOPNET_8678`).

- **ThingSpeak Configuration:**

- `server`: ThingSpeak API endpoint (`http://api.thingspeak.com`).
- `channelID`, `fieldNumber`, `writeAPIKey`, `readAPIKey`: Parameters for reading weather data (`field3`) and writing soil moisture data.

- **Pin Definitions:**

- `soilPin` (A0): Analog input for the soil moisture sensor.
- `pumpLedPin` (D1): LED indicating pump status.
- `pumpRelayPin` (D2): Relay controlling the water pump.
- `rainLedPin` (D6): LED indicating expected rain.
- `statusLed` (D7): LED for general system status.

- **Constants:**

- `dryThreshold` (600): Threshold for dry soil (sensor value > 600).
- `manualDuration` (10000 ms): Duration for manual pump activation.
- `checkInterval` (30000 ms): Interval for checking soil moisture and conditions.

2.2 Setup Function

The `setup()` function initializes the system:

1. **Serial Communication:**

- Starts serial communication at 115200 baud for debugging.

2. **Pin Configuration:**

- Sets `pumpLedPin`, `pumpRelayPin`, `rainLedPin`, and `statusLed` as outputs, initially set to LOW (off).

3. **Wi-Fi Connection:**

- Connects to the specified Wi-Fi network using `WiFi.begin()`.
- Logs connection progress to the Serial Monitor and Blynk (virtual pin V10).

4. **Blynk Initialization:**

- Connects to the Blynk server using the provided credentials.

5. **Time Synchronization:**

- Configures NTP servers (`pool.ntp.org`, `time.nist.gov`) for accurate time.
- Waits until time is synchronized (`epoch > 100000`).

2.3 Loop Function

The `loop()` function handles the main system logic:

1. **Blynk Execution:**

- Calls `Blynk.run()` to process Blynk events (e.g., manual control input).

2. **Manual Override Check:**

- If `manualOverride` is active, checks if the manual pump duration (10 seconds) has elapsed.
- If elapsed, turns off the pump and LED, resets `manualOverride`, and logs the event.

3. Automatic Mode (every 30 seconds, if no manual override):

- Reads soil moisture from `soilPin` (A0).
- Logs the value to Serial and Blynk (V10).
- Sends the value to ThingSpeak for logging.
- Retrieves the current hour via NTP.
- **If soil is dry** (`soilValue > 600`):
 - Fetches rain forecast from ThingSpeak.
 - If rain is expected (`rainExpected == 1`):
 - Turns off pump and pump LED, turns on rain LED, logs the event.
 - If no rain is expected:
 - Checks if the current time is within irrigation windows (5:00–9:00 AM or 6:00–8:00 PM).
 - If within the window, activates the pump and pump LED, logs the event.
 - If outside the window, keeps the pump off, turns on rain LED (indicating "not the right time"), logs the event.
- **If soil is humid** (`soilValue <= 600`):
 - Keeps the pump and all LEDs off, logs the event.

2.4 Blynk Callback (`BLYNK_WRITE(V0)`)

- Triggered when the Blynk app sends a value to virtual pin V0 (manual control button).
- If `value == 1`:
 - Activates the pump and pump LED, logs the event, sets `manualOverride` to true, and records the start time.
- If `value == 0`:
 - Turns off the pump and LEDs, resets `manualOverride`, logs the event.

2.5 Helper Functions

- **logToTerminal(String msg):**
 - Logs messages to the Serial Monitor and Blynk (V10) for debugging and user feedback.
- **sendSoilToThingSpeak(int value):**
 - Sends the soil moisture value to ThingSpeak's field1 using an HTTP GET request.
 - Logs success or failure.
- **getRainForecast():**
 - Fetches the latest weather data from ThingSpeak's field3 using an HTTP GET request.
 - Returns 1 if rain is expected, 0 otherwise.
 - Logs the request URL, response, and any errors.

3. Key Features

- **Automated Irrigation:** Combines soil moisture, weather, and time data to irrigate only when necessary.
- **Manual Control:** Allows users to override automation via the Blynk app.
- **Real-Time Monitoring:** Logs soil moisture to ThingSpeak and displays status on Blynk.

- **Feedback:** Uses LEDs for visual indication of pump, rain, and system status.
- **Reliability:** Handles Wi-Fi disconnections and HTTP errors gracefully.

4. Limitations and Potential Improvements

- **Pin Mismatch:** The code uses `pumpRelayPin` (D2), while the report specifies D0. This should be corrected to ensure consistency with hardware.
- **Error Handling:** Limited retry logic for failed HTTP requests or Wi-Fi issues.
- **LED Usage:** The `statusLed` (D7) is defined but rarely used; its role could be expanded for better system feedback.
- **Future Enhancements:**
 - Add temperature/light sensors for more precise irrigation.
 - Implement SMS/Telegram alerts for anomalies.
 - Use solar power for sustainability.
 - Analyze long-term ThingSpeak data for irrigation optimization.

5. Conclusion

The code implements a robust, low-cost IoT solution for smart irrigation, integrating soil moisture sensing, weather forecasts, and time-based scheduling. It leverages the ESP8266's Wi-Fi capabilities and platforms like Blynk and ThingSpeak for remote control and monitoring, making it suitable for small to medium-scale farming.